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Thread class methods (Part 1) in Java

=> Thread class constructors :-

1. public Thread() { - }
2. public Thread(Runnable target) { - }
3. public Thread(String name) { - }
4. public Thread(Runnable target, String name) { - }
5. public Thread(ThreadGroup group, Runnable target) { - }
6. public Thread(ThreadGroup group, String name) { - }
7. public Thread(ThreadGroup group, Runnable target, String name) { - }
8. public Thread(ThreadGroup group, Runnable target, String name, long stackSize) { - }

=> Thread class methods :-

1. Simple methods :-

➔ run() - this method contains the thread task
➔ start() - this method is used to create thread

- ➔ currentThread() - this method returns the reference of current running thread
- ➔ isAlive() - this method is used to check wether the thread is running or not

2. Naming Methods :-

- ➔ getName() - this method is used to get the current running thread name
- ➔ setName(String name) - this method is used to set the name of current running thread

3. Daemon thread methods :

- ➔ isDaemon() - this method is used to check wether the thread is daemon thread or not
- ➔ setDaemon(boolean b) - this method is used to create daemon thread

=> Daemon Threads :-

- ➔ Daemon threads are the threads which are executed in the background of another thread
- ➔ For example : Garbage collector, finalizer, spelling checker in word etc
- ➔ Daemon thread is used to provide the service to the other threads
- ➔ Daemon thread methods :
  - isDaemon()
  - setDaemon(boolean b)

➔ Note :

1. We cannot create the running thread as daemon thread. If we try to create daemon thread after thread has started then it will throw an exception saying "java.lang.IllegalThreadStateException"
2. We cannot create main thread as daemon thread because JVM starts the main thread before we create it as daemon thread

3. Daemon thread life depends on another thread in which it is running
4. Daemon thread inherits the properties/nature from its parent thread
5. JVM can stop the daemon thread but it cannot stop the normal thread
6. It is recommended to make the lowest priority of daemon thread

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⇨ Thread class methods diagram :

"Thread" class methods (Part 1)

```
public class Thread implements Runnable
{
    public void run() { - }
    public synchronized void start() { - }
    public static native Thread currentThread();
    public final native boolean isAlive();

    public final String getName() { - }
    public final synchronized void setName(String name) { - }

    public final boolean isDaemon() { - }
    public final void setDaemon(boolean on) { - }

    public final int getPriority() { - }
    public final void setPriority(int newPriority) { - }

    ----- (many more methods)
}
```

→ Basic Methods

→ Naming Methods

→ Daemon Thread Methods

→ Priority Based Methods

"Thread" class methods (Part 2)

```
public class Thread implements Runnable
{
    ----- (many more methods)

    public static native void sleep(long millis) throws InterruptedException;
    public static native void yield();
    public final void join() throws InterruptedException { - }
    public final void suspend() { - }
    public final void resume() { - }
    public final void stop() { - }
    public void destroy() { - }

    public void interrupt() { - }
    public boolean isInterrupted() { - }
    public static boolean interrupted() { - }
}
```

→ Prevent Thread Execution Methods

→ Depricated methods

→ Interrupting a thread Methods

Company Links & Contacts

Company Name: Smart Programming (+91 62838-30308)

Address : Chandigarh & Mohali (Punjab), India

Websites: https://www.smartprogramming.in/

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